Murmurs, Fetal Circulation, and Heart Defects

From Chapter 38 (your pasted text). Fetal circulation is from Section 12 of your study guide, because it isn't in the pasted chapter.

1. Innocent murmurs

What they are: the sound of normal blood flowing through a normal heart. Nothing is broken. **Up to 80% of children** have one at some point.

Innocent = the 7 S's (Box 38.2):

- **Soft:** grade I–III, and localized (it doesn't spread)
- **Systolic:** except the venous hum, which is continuous
- **Short:** peaks in the first half of systole
- **Single:** no clicks, snaps, or extra sounds
- **Small area:** heard best at the LLSB or pulmonic area, and rarely radiates
- **Sweet:** musical, vibratory, sometimes blowing (never harsh)
- **Sensitive:** changes with position, and may disappear with Valsalva or gentle pressure on the neck vein

Also expected with an innocent murmur:

- Normal vital signs, a normal ECG, and a healthy child
- It may come and go between visits
- It gets **louder with fever, anemia, exercise, anxiety, or pain**, because these increase cardiac output

Teaching for families: the heart is normal. The murmur may come and go. **No activity limits and no special precautions.**

The 4 innocent murmurs (Table 38.5)

Murmur	Age	Where	Clues	Can be mistaken for
Still	2–6 yrs	LLSB/apex, doesn't radiate	"Twangy string," vibratory, **louder lying down**, softer standing, grade II	Small VSD, IHSS
Pulmonary flow murmur of childhood	8–14 yrs, thin-chested	Pulmonic area	Soft, blowing, **normal split S2**, no click or thrill, grade I–II	ASD, pulmonic stenosis
Pulmonary flow murmur of infancy (peripheral pulmonic stenosis)	Newborns, especially preemies	LUSB → spreads to **both axillae and the back**	Soft, **gone by 6 months**	Supravalvular PS or AS
Venous hum	2–8 yrs	Upper chest under the clavicles, when upright	Continuous swishing that **disappears when lying down, turning the head, or pressing the neck vein**	**PDA**

🧠 **"Still = Strings, Supine louder." "Hum goes Home when you lie down."**

Red flags: possibly PATHOLOGIC (refer to cardiology)

- **Any diastolic murmur**
- A thrill, which means grade IV or louder
- A pansystolic (holosystolic) murmur

- A continuous murmur that **can't be made to go away**
- Clicks, opening snaps, or an S4
- Fixed split S2, or a loud (accentuated) S2
- A **harsh** murmur, or one that **doesn't change with position**
- A child with a syndrome linked to heart defects (for example, **trisomy 21**)

> 🧠 **"Diastolic is always Dangerous." "Loud ≠ bad, soft ≠ safe."** A small VSD can be loud, and a large defect may make no murmur at all.

2. Grading murmurs (Levine I–VI)

Grade	What it sounds like
I	Barely audible. You hear it faintly only after listening carefully.
II	Soft, but easy to hear
III	Moderately loud, **no thrill**
IV	Loud, heard over a wide area, **palpable thrill** (the thrill starts here)
V	Heard with the stethoscope **barely touching** the chest, plus a thrill
VI	Heard **without** the stethoscope on the chest (rare)

> 🧠 **"4 = you can FEEL it. 5 = barely touching. 6 = not touching."**

Key facts:

- **The grade does NOT tell you how severe the problem is.**
- The grade can change when the child moves from lying down to sitting.

To fully describe a murmur, include: timing (systolic, diastolic, or continuous), location, radiation, quality, duration, pitch, and grade.

Timing:

- **Systolic** (between S1 and S2): most murmurs, and every innocent murmur
- **Diastolic** (between S2 and S1): **always pathologic**
- **Continuous** (through both systole and diastole): most often a **PDA**. Tell it apart from a venous hum.

Where to listen:

- **URSB:** aortic stenosis click, venous hum
- **ULSB:** pulmonic stenosis click, ASD, PDA, pulmonary flow murmurs
- **LLSB:** VSD, Still, HCM, tricuspid regurgitation
- **Apex:** mitral regurgitation, the MVP click
- **Erb point:** aortic ejection click

3. Fetal circulation (study guide, Section 12)

Before birth:

- The **placenta** does the lungs' job, so blood mostly skips the lungs.
- The umbilical **vein** brings oxygenated blood **to** the baby.
- The umbilical **arteries** take blood **back** to the placenta.

The 3 shunts:

Shunt	What it does	Mnemonic
Ductus venosus	Skips the liver	"Venosus skips the liVer"
Foramen ovale	Right atrium → left atrium	"Ovale = Open door between the atria"
Ductus arteriosus	Pulmonary artery → aorta (skips the lungs)	"Arteriosus connects the 2 Arteries"

At birth:

1. The lungs expand, so **pulmonary resistance drops**.
2. The cord is clamped, so **systemic resistance rises**.
3. Left atrial pressure rises, which pushes the **foramen ovale** closed.
4. More oxygen and less prostaglandin make the **ductus arteriosus** constrict.

From Chapter 38: the ductus closes **functionally in 12–72 hours** and **permanently in 2–3 weeks** in term babies.

> 🧠 **"Prostaglandin keeps it Patent. Indomethacin Interrupts (closes) it."**

Why this matters: after birth, lung resistance keeps falling over the first **2–8 weeks**. More blood then crosses a VSD to the lungs, which is why a VSD murmur and heart failure appear **weeks after a normal newborn exam**.

4. ACYANOTIC defects (Fig 38.6)

A. Left-to-right shunts: too much blood goes to the lungs

> 🧠 **"The D's flood the lungs."** ASD, VSD, PDA, and AVSD all end in D.

ASD (hole between the atria)

- Most common type: **secundum**, near the foramen ovale. **Primum** is lower and seen in **Down syndrome**.
- Often no symptoms. Some children are thin, tire easily, or get frequent colds or pneumonia.
- The murmur often isn't noticed until **2–3 years** old.
- **Wide, fixed split S2.** There is also a soft systolic murmur at the pulmonic area, caused by extra flow across the **pulmonary valve**, not by the hole itself.
- ECG: right axis deviation and an **rSR′ pattern in V1**.
- **Treatment:** small holes may close on their own. Device closure if the child weighs **more than 15 kg**, then **aspirin for 6 months** and endocarditis prophylaxis for 6 months. No exercise limits.
- **If untreated:** right-sided heart failure, clots crossing the hole to the body (paradoxical emboli), pulmonary hypertension.

> 🧠 **"ASD = Always Split, Doesn't change."**

VSD (hole between the ventricles)

- **The most common** heart defect. The most common type is **perimembranous**.
- **No murmur at birth.** As lung resistance falls (2–8 weeks), the loud murmur and heart failure appear.
- **Small VSD:** harsh, high-pitched holosystolic murmur at the **LLSB**; the child is otherwise normal. Most close by about **age 4**. **No endocarditis prophylaxis.**
- **Large VSD:** low-pitched holosystolic murmur, a diastolic rumble at the apex, a thrill, and heart failure with S3 or S4.
- A **higher-pitched murmur over time means the hole is getting smaller.**
- **Heart failure care:** digoxin, diuretics, ACE inhibitors, high-calorie formula (24/27/30 kcal/oz), and an NG tube if needed.
- Device closure if the child weighs **more than 6 kg**.

> 🧠 **"VSD = Very loud. Small hole, loud howl."**

AV septal defect (AV canal)

- **Complete** = primum ASD + **one shared AV valve** + inlet VSD.
- Strongly linked to **Down syndrome**. ECG: **superior axis**. Repaired **before 6 months** of age.
- **Watch out:** some babies with Down syndrome have a **loud single S2 and almost no murmur**. That means pulmonary hypertension, not a healthy heart.

PDA (open ductus arteriosus)

- Seen in **45–80% of preemies under 1750 g**. Girls are affected 3 times as often as boys.
- Right after birth the murmur is soft and systolic. After a few weeks it becomes a **continuous "machinery" murmur** under the left clavicle, with **bounding pulses and a wide pulse pressure**.
- **Indomethacin or ibuprofen closes it in PREEMIES ONLY.** It doesn't work in term or older infants.
- Other options: coils or plugs placed in the cath lab (after about 8 months of age), or surgical ligation (no heart-lung bypass needed).

> 🧠 **"PDA = Machinery, Wide, Bounding."**

B. Obstructive defects: blood is blocked, but stays pink at first

Coarctation of the aorta (narrowed aorta)

- **High BP in the arms and low BP in the legs.** Arm pulses are bounding; **femoral pulses are weak or delayed**.
- Older child: headaches and leg pain with exercise. **Rib notching** on x-ray.
- Newborn: crashes when the **ductus closes**, so **PGE1** is given. Infant: fast breathing, poor feeding, cool legs.
- Linked to **Turner syndrome**. Most also have a bicuspid aortic valve.
- After repair: check BP for life. Sports are OK if the **arm-leg BP difference is under 20 mm Hg**.

> 🧠 **"Strong upstairs, weak downstairs."**

Aortic stenosis (narrowed aortic valve)

- Usually a **bicuspid** valve. Thickens the **left ventricle**. Boys are affected 4 times as often as girls.
- Harsh grade III–IV murmur at the **URSB that radiates to the neck**, a thrill at the suprasternal notch, and a **narrow** pulse pressure.
- Risks: exertional chest pain, fainting, **sudden death**.
- **Sports:** mild = all sports; moderate = low-intensity only (golf, bowling); severe = no competitive sports.
- A mechanical replacement valve means **warfarin**.

> 🧠 **"Aortic = Left, and into the neck."**

Pulmonic stenosis (narrowed pulmonary valve)

- Thickens the **right ventricle**. Usually no symptoms, and found on a routine exam.
- Harsh murmur at the **ULSB that spreads to the back and lungs**. The click **gets quieter when the child breathes in**.
- A **critical** case in a newborn causes cyanosis.
- Balloon valvuloplasty if the pressure gradient is **over 50 mm Hg**.
- Linked to **Noonan and Williams** syndromes.

> 🧠 **"Pulmonic = Pushes the Right side."**

5. CYANOTIC defects (right-to-left flow or mixing)

General points:

- Cyanosis becomes **visible at about 85%** oxygen saturation. It's easier to see with polycythemia, and harder with anemia or fetal hemoglobin.
- **Polycythemia** helps carry oxygen but raises the **risk of stroke**, so keep these children hydrated.
- **Hyperoxia test:** if 100% oxygen pinks the baby up, the problem is the **lungs**. If the baby stays blue, it's the **heart**.

> 🧠 **The 5 T's, plus HLHS and critical PS:** Truncus (1 trunk), Transposition (2 vessels switched), Tricuspid atresia (3), Tetralogy (4 defects), TAPVR (5 letters)

Tetralogy of Fallot (the most common cyanotic defect)

- **PROV:** Pulmonary stenosis, RV hypertrophy, Overriding aorta, VSD
- Cyanosis gets **worse over the first months**. "Pink tets" have mild blockage and little cyanosis.
- **Tet spells** at 2–4 months, triggered by **crying, feeding, or pooping**. Treatment: **knee-chest position**, soothe the baby, oxygen, and morphine. Older children **squat**, which does the same thing.
- Harsh murmur at the left mid-to-upper sternal border. **Boot-shaped heart** on x-ray.

- Repaired in infancy. Follow for life for pulmonary valve leak, arrhythmias, and a widening QRS (sudden death risk).
- Linked to **22q11 (DiGeorge)** and Down syndrome.

> 🧠 **"Tet spell? Tuck like a frog."**

d-Transposition of the great arteries

- The aorta comes from the right ventricle and the pulmonary artery from the left. Blood runs in **two separate circles**.
- **Blue within the first hour to day.** Often a **big baby** (infant of a diabetic mother). **Loud single S2. Egg-shaped heart** on x-ray.
- Treatment: **PGE1**, a **balloon atrial septostomy** to mix the blood, then an **arterial switch in the first days**. Endocarditis prophylaxis for life.
- **Without treatment, 50% die by 1 month and 90% by 1 year.**

> 🧠 **"Open (PGE1), Mix (septostomy), Switch."**

Tricuspid atresia (single ventricle)

- **No valve between the right atrium and right ventricle.** Blood has to cross an ASD to the left side. The right ventricle is small.
- **50% also have TGA.**

> 🧠 **"Front door locked? Take the side door."**

Hypoplastic left heart syndrome (single ventricle)

- The left heart is tiny. The **body's blood supply depends on the ductus**, so the baby **goes into shock as the ductus closes**.
- Linked to **Turner and Jacobsen** syndromes.

Staged surgery for single-ventricle defects:

1. Newborn: a shunt
2. **4–6 months: Glenn** (SVC connected to the pulmonary artery)
3. **2–4 years: Fontan** (IVC connected to the pulmonary artery)
4. Transplant if the stages fail

- Complications: protein-losing enteropathy, arrhythmias, clots and stroke, low exercise tolerance.
- Endocarditis prophylaxis **while the child is still cyanotic**.

> 🧠 **"Superior first, Inferior last"** (S comes before I in the alphabet).

Truncus arteriosus

- One single great artery feeds both the body and the lungs. Linked to **DiGeorge**. **Critical pulmonic stenosis** is also on the cyanotic list.

One-glance summary

ACYANOTIC: too much to the lungs	ACYANOTIC: blocked	CYANOTIC
ASD, VSD, PDA, AV canal	Coarctation, aortic stenosis, pulmonic stenosis	TOF, TGA, tricuspid atresia, HLHS, truncus, critical PS
Pink, but may develop heart failure	Pink at first, can crash	Blue